6 Finite Beams
Radiation, Diffraction, and Scattering

There are many advantages in using plane wave solutions; they are conceptually simple and greatly facilitate the mathematics. However, in practice, ultrasonics always involves the use of finite size transducers, hence finite beams. An immediate consequence is that diffraction effects must be considered. Furthermore, the consequences for focusing, imaging, leaky waves, and scattering from various sizes and shapes of objects, among other issues, must be addressed.

One of the many issues of this chapter will be the treatment of diffraction effects. For simplicity, we deal with scalar theory for a fluid medium although the results can be directly extended to a solid medium. We start with radiation from a point source and then extend the discussion to radiation by a circular piston. This is a classic problem in ultrasonics and the results give general guidelines for the emission of ultrasonic waves from a piezoelectric transducer. Following that, an outline is provided for the scattering of ultrasonic waves by circular and cylindrical obstacles. Finally, the main issues involved in focused ultrasonic waves, acoustic radiation pressure and the Doppler shift, are addressed.

6.1 RADIATION

6.1.1 Point Source

It is convenient to write the wave equation in a totally general form for a fluid:

$$\frac{\partial^2 p}{\partial t^2} = V_0^2 \quad {}^2 p \tag{6.1}$$

Since p is a scalar, we can use any coordinate system and employ the appropriate form of ∇^2.

Clearly a spherical coordinate system is best suited for problems dealing with a point source. Doing the transformation $(x, y, z) \rightarrow (r, \theta, \psi)$, we have

$$\begin{aligned} x &= r\sin\theta\cos\psi \\ y &= r\sin\theta\sin\psi \\ z &= r\cos\theta \end{aligned} \tag{6.2}$$

and

$$= \quad {}^2 = \frac{\partial^2}{\partial r^2} + \frac{2}{r}\frac{\partial}{\partial r} + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta} \quad \sin\theta\frac{\partial}{\partial\theta} \quad + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\psi^2} \tag{6.3}$$

Substituting into Equation 6.1 and noting that for spherical waves the pressure is independent of θ and ψ:

$$\frac{\partial^2 p}{\partial t^2} = V_0^2 \quad \frac{\partial^2 p}{\partial r^2} + \frac{2}{r}\frac{\partial p}{\partial r} \tag{6.4}$$

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Hence,

$$\frac{\partial^2 p}{\partial t^2} = V_0^2 \quad \frac{1}{r}\frac{\partial^2 (rp)}{\partial r^2} \tag{6.5}$$

Since r and t are independent variables, this can be rewritten

$$\frac{\partial^2 (rp)}{\partial t^2} = V_0^2 \frac{\partial^2 (rp)}{\partial r^2} \tag{6.6}$$

with solutions

$$rp = f_{\mathrm{d}}(V_0 t - r) + f_{\mathrm{c}}(V_0 t - r) \tag{6.7}$$

$$p = \frac{f_{\mathrm{d}}(V_0 t - r)}{r} + \frac{f_{\mathrm{c}}(V_0 t - r)}{r} \tag{6.8}$$

f_{d}, the solution for diverging waves, will be mainly useful for radiation problems, while f_{c}, the converging solution, will be appropriate for focused spherical waves. The diverging solution will be treated explicitly in what follows.

As for plane waves, it will useful to develop relations between displacement, particle velocity, dilatation, and pressure. From Newton's law

$$-\frac{\partial p}{\partial r} = \rho_0 \frac{\partial^2 u_{\mathrm{r}}}{\partial t^2} \tag{6.9}$$

where u_{r} is the radial particle displacement.

This can be integrated to give

$$v_{\mathrm{r}} = \frac{\partial u_{\mathrm{r}}}{\partial t} = -\frac{1}{\rho_0}\int \frac{\partial p}{\partial r}\mathrm{d}t \tag{6.10}$$

or in complex form

$$v_{\mathrm{r}} = -\frac{1}{j\omega\rho_0}\frac{\partial p}{\partial r} \tag{6.11}$$

Finally, the displacement is

$$u = \int v\mathrm{d}t = \frac{v}{j\omega} = \frac{1}{\rho_0\omega^2}\frac{\partial p}{\partial r} \tag{6.12}$$

For harmonic solutions at frequency ω,

$$p = \frac{A}{r}\exp j(\omega t - kr) \tag{6.13}$$

Hence,

$$S = -\frac{p}{\rho_0 V_0^2} \tag{6.14}$$

$$u_{\mathrm{r}} = -\quad \frac{1}{r} + jk \quad \frac{p}{\rho_0\omega^2} \tag{6.15}$$

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$$v_r = \frac{1}{r} + jk \quad \frac{p}{j\omega\rho_0} \tag{6.16}$$

Contrary to the case for plane waves, the particle velocity is in general out of phase with the pressure while the displacement always lags the pressure by $\pi/2$.

The specific acoustic impedance is given by

$$Z = \frac{p}{v} = \frac{j\omega\rho_0}{\left(\frac{1}{r} + jk\right)} = \frac{\rho_0 V_0 (kr + j)}{1 + k^2 r^2} \tag{6.17}$$

$$= \rho_0 V_0 \frac{k^2 r^2}{1 + k^2 r^2} + j\rho_0 V_0 \frac{kr}{1 + k^2 r^2} \tag{6.18}$$

where the phase angle between real and imaginary parts is $\tan\theta = 1/kr$ or

$$\cos\theta = \frac{kr}{\sqrt{1 + k^2 r^2}} \tag{6.19}$$

The modulus of the acoustic impedance

$$|Z| = \left|\frac{p}{v}\right| = \rho_0 V_0 \frac{kr}{\sqrt{1 + k^2 r^2}} = \rho_0 V_0 \cos\theta \tag{6.20}$$

which approaches the value for plane waves for $kr \gg 1$. This is as expected because far from the source the spherical wave approximates a plane wave.

By definition $Z \equiv p/v$, the particle velocity can be expressed in terms of the impedance as

$$v = \frac{A}{rZ} \exp j(\omega t - kr) \tag{6.21}$$

The intensity I of a spherical wave is by definition the average rate of work done per area on the surrounding medium. For a cycle of period T

$$I \equiv \frac{\int_0^T pv\mathrm{d}t}{T} \tag{6.22}$$

Using the real part of p and v and the previous results for the phase angle

$$\begin{aligned} I &= \frac{1}{T}\int_0^T p_0 \cos(\omega t - kr) v_0 \cos(\omega t - kr - \theta)\mathrm{d}t \\ &= \frac{p_0 v_0 \cos\theta}{2}, \quad \text{where } p_0 = \frac{A}{r} \end{aligned} \tag{6.23}$$

Using $v_0 \cos\theta = p_0/\rho_0 V_0$ from Equation 6.20

$$I = \frac{p_0^2}{2\rho_0 V_0} \tag{6.24}$$

It is now possible to formulate the sound field associated with spherical waves. Assuming a spherical source of radius a immersed in a fluid, the radial velocity at a point on the surface is given by

$$v = v_0 \mathrm{e}^{j\omega t} \tag{6.25}$$

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For small amplitudes, the boundary condition is continuity of the radial velocity. From the previous results,

$$\frac{A}{aZ_a}\exp j(\omega t - ka) = v_0 \exp(j\omega t)$$

so that

$$A = av_0 Z_a \exp j(ka) \approx j\rho_0 V_0 ka^2 v_0, \quad \text{for } ka \ll 1 \tag{6.26}$$

This gives the desired result for a small (point) source

$$p = \frac{j\rho_0 V_0 ka^2 v_0}{r}\exp j(\omega t - kr) \tag{6.27}$$

6.1.2 Radiation from a Circular Piston

The result for a circular piston will, of course, be much more complex, and much more difficult to calculate, than that from a point source. The basic principle is simple; each point of the source can be treated as a point-like source, emitting spherical waves as given by Equation 6.27 at distances far enough away from the source. Then by the Huygens principle, these various contributions can be summed, taking into account the amplitude and the phase from each contribution. In practice, analytical results are difficult to obtain, even for the simplest cases, leading to a choice between numerical calculation and approximate solutions. The latter approach is chosen here.

The circular piston radiator is an important example in ultrasonics as it is about the simplest approximation that can be made for radiation into an infinite medium. It is also important in audio acoustics in the theory of loudspeakers. The assumption will be made that it is mounted inside an infinite baffle, so that sound is only radiated in the forward direction. The geometry is shown in Figure 6.1, where it is assumed that the transducer is excited with uniform particle velocity across its face.

From Equation 6.27, each infinitesimal source element area $\overrightarrow{dA}$ on the piston produces a differential pressure dp at a point of observation at a distance of $\vec{r}$ given by

$$dp = \frac{j\rho_0 V_0 k}{2\pi r}(\vec{v}\cdot\overrightarrow{dA})\exp j(\omega t - kr\ ) \tag{6.28}$$

Since the motion of each element is normal to the surface, dp can be written as

$$dp = \frac{j\rho_0 V_0 k}{2\pi r} v dA\ \exp j(\omega t - kr\ ) \tag{6.29}$$

The total pressure p at the point (r, θ) is the integral of dp over the full radiator surface.

From elementary geometry

$$r\ = (r^2 + \sigma^2 - 2r\sigma\sin\theta\cos\psi)^{\frac{1}{2}} \tag{6.30}$$

However, the resulting expression for dp is not integrable and so approximations have to be made. We first treat the far-field or Fraunhofer limit, where $r \gg a$.

6.1.2.1 Fraunhofer (Far-Field) Region

For $r', r \gg a$, this can be expanded in a Taylor's series with the first two terms

$$r\ = r - \sigma\sin\theta\cos\psi$$

Since the distance between two neighboring points is crucial for an accurate calculation of the phase difference between pressure waves emitted from them, both these terms must be retained for the phase. For the amplitude, $r' \approx r$ is sufficient.

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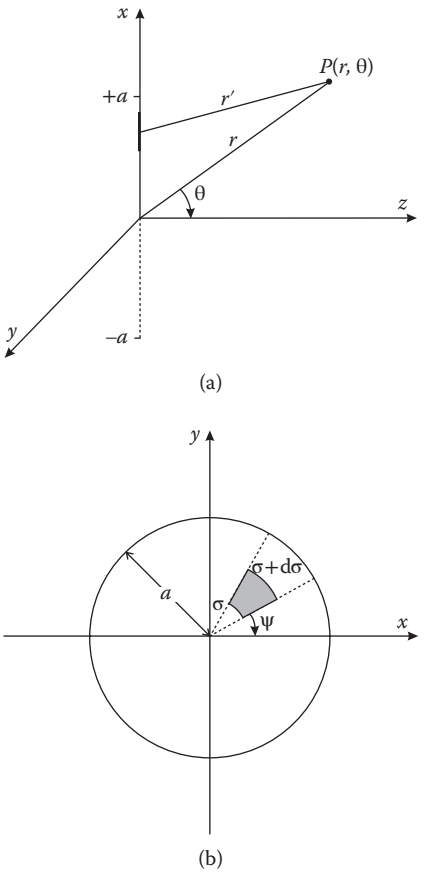

FIGURE 6.1 Geometrical variables used for the calculation of the pressure distribution for a plane piston circular radiator. (a) Axial view. (b) Radial coordinates in the plane of the radiator.

We then have

$$p = \frac{j\rho_0 V_0 k}{2\pi r} v_0 e^{j(\omega t - kr)} \int_0^a \sigma d\sigma \int_0^{2\pi} e^{jk\sigma \sin\theta \cos\psi} d\psi \tag{6.31}$$

The second integral (over ψ) can be expanded as a power series and integrated to give $2\pi J_0(k\sigma \sin\theta)$ (see Appendix A). The second integral can be obtained from

$$\int x J_0(x)\, dx = x J_1(x)$$

so that

$$2\pi \int_0^a \sigma J_0(k\sigma \sin\theta)\, d\sigma = 2\pi a^2 \frac{J_1(ka \sin\theta)}{ka \sin\theta}$$

so that finally

$$p = \frac{j\rho_0 V_0 k a^2 v_0}{2r} e^{j(\omega t - kr)} \frac{2J_1(ka \sin\theta)}{ka \sin\theta} \tag{6.32}$$

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The term in brackets is known as the directivity function (DF) as it gives the variation of pressure with direction. Numerical values are tabulated in Appendix A and the function is plotted in Figure 6.2.

An approximate form for the DF can be obtained by expanding $J_1(x)$, yielding

$$\frac{2J_1(x)}{x} \approx 1 - \frac{x^2}{8}$$

In particular, for points along the axis $x = 0$, then DF = 1 and the result for p is identical in form to that of a point-like source of area πa^2.

The first zero θ_1 of jinc x occurs at $ka \sin\theta = 3.83$; hence,

$$\sin\theta_1 = \frac{3.83}{ka} = 0.61\frac{\lambda}{a} \tag{6.33}$$

which gives a measure of the angular half width of the principle lobe of the acoustic pressure. By the same token, the first sidelobe is included between the angles θ_1 and θ_2, where

$$\sin\theta_2 = \frac{7.02}{ka} = 1.12\frac{\lambda}{a} \tag{6.34}$$

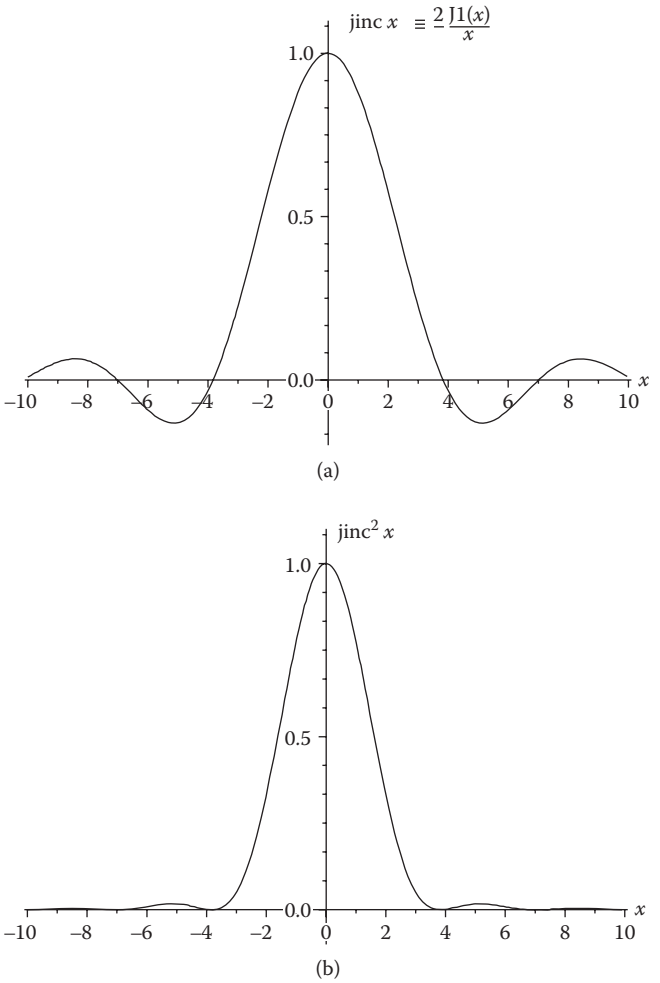

FIGURE 6.2 Directivity function jinc x for the circular radiator. (a) Pressure. (b) Intensity.

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In this way, one can identify a whole series of lobes, on either side of the main lobe, called the sidelobes. These sidelobes are undesirable for two reasons. The main objective of an acoustic radiator is to produce a narrow collimated beam of acoustic energy to be used in some application, for example, imaging or nondestructive testing. The sidelobes represent energy lost from the main beam, which is of course undesirable. If the sidelobes are big enough, they can interfere with information obtained from the main beam, which is also unwanted. Hence, an important part of the design of acoustic radiators involves sidelobe reduction.

An alternative and more efficient way to present the sidelobes is by the use of polar plots, as shown in Figures 6.3 and 6.4, for several different frequencies using both decibel and linear scales. It is seen that for $ka \gg 1$, there are many sidelobes. As ka decreases, the number is reduced and for $ka \ll 1$ there is really only the main lobe, for which the $DF \oplus 1$. In this case, the axial intensity is given by

$$I_0 = \frac{\rho_0 V_0 k^2}{8\pi r^2} v_0^2 A \tag{6.35}$$

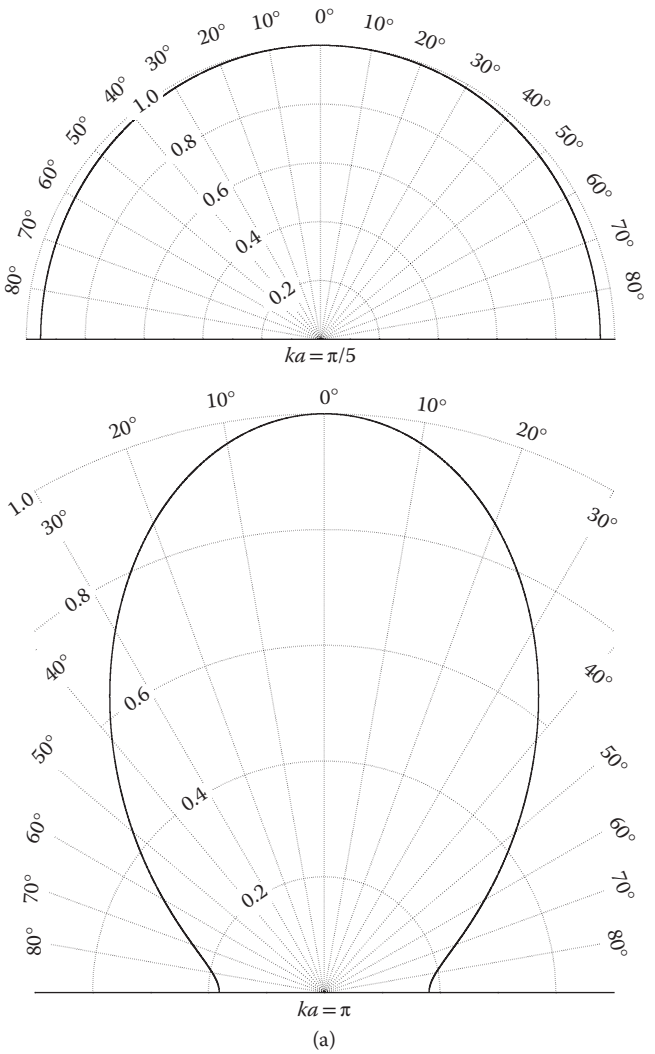

FIGURE 6.3 (a) Polar diagrams (linear scale) for circular radiators with radius/wavelength ratios of (top) 0.1 and (bottom) 0.5. (b) Polar diagrams (linear scale) for circular radiators with radius/wavelength ratios of (top) 2 and (bottom) 10.

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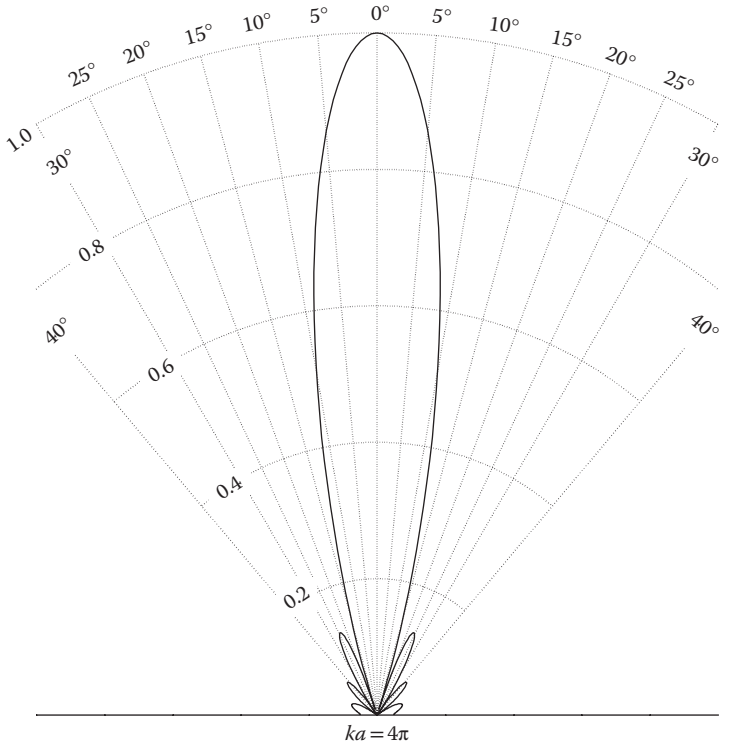
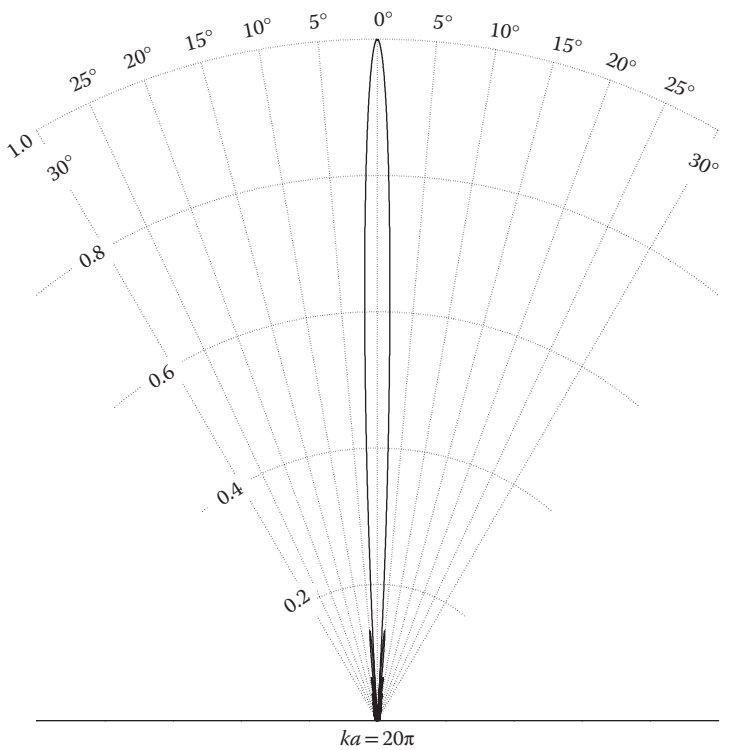


$ka = 4\pi$



$ka = 20\pi$

(b)

FIGURE 6.3 (*continued*)

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where A is the area of the piston. Deep in the far-field region $I_0 \propto 1/r^2$, which is physically reasonable as far from the radiator, the latter looks like a point-like source.

6.1.2.2 Fresnel (Near-Field) Approximation

This is the opposite limiting case where the observation point is near the transducer; a quantitative criteria will be given at the end of the section. As the name implies, the situation closely resembles that of Fresnel diffraction in optics.

Analytical solutions are only available on the axis, where $r' = (r^2 + \sigma^2)^{1/2}$. Hence, from Equation 6.28

$$\mathrm{d}p = \frac{j\rho_0 V_0 k}{2\pi} v_0 \mathrm{e}^{j\omega t} \frac{\mathrm{e}^{-jk\sqrt{r^2+\sigma^2}}}{\sqrt{r^2+\sigma^2}} \mathrm{d}A \tag{6.36}$$

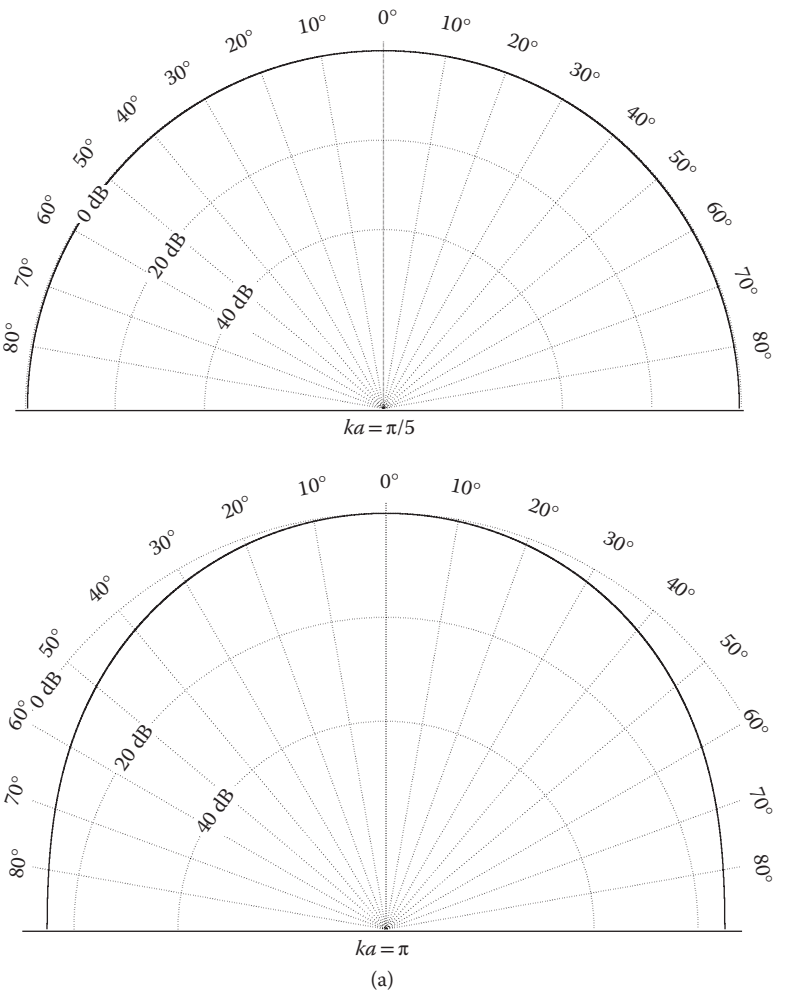

FIGURE 6.4 (a) Polar diagrams (log scale) for circular radiators with radius/wavelength ratios of (top) 0.1 and 0.5 (bottom). (b) Polar diagrams (log scale) for circular radiators with radius/wavelength ratios of (top) 2 and (bottom) 10.

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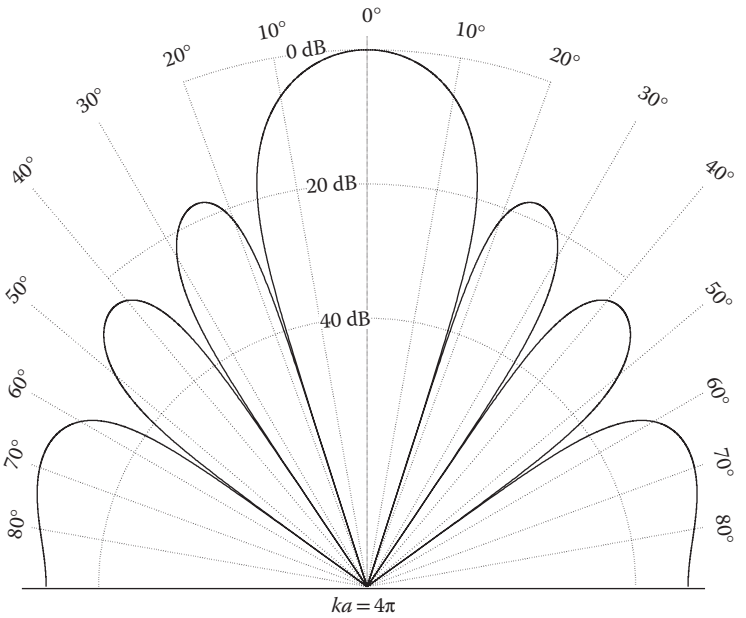

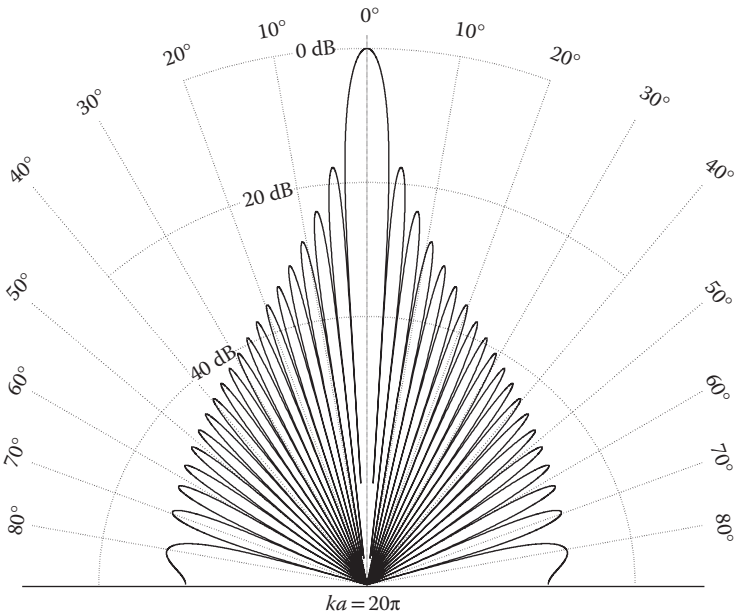

(b)

FIGURE 6.4 (*continued*)

After integration

$$p = \rho_0 V_0 v_0 e^{j\omega t}\left(e^{-jk\sqrt{r^2+\sigma^2}} - e^{-jkr}\right)dA$$

with real part

$$p = \rho_0 V_0 v_0 a^2 \quad 2 - 2\ \cos k\left(\sqrt{r^2 + z^2} - z\right)^{\frac{1}{2}} \tag{6.37}$$

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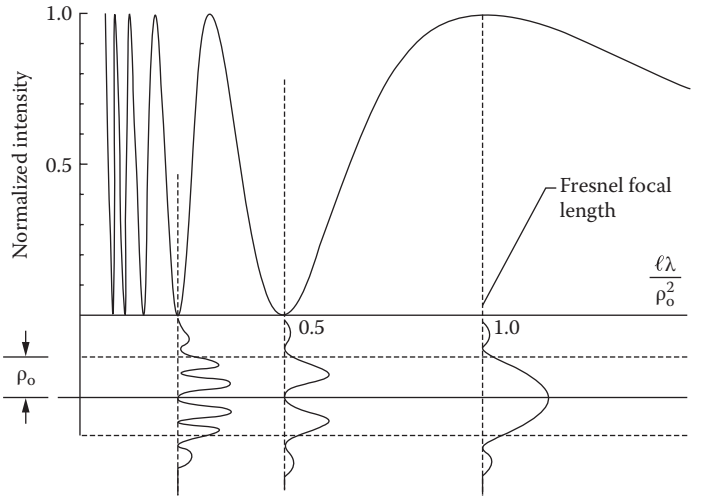

FIGURE 6.5 Axial intensity distribution produced by a circular transducer of radius ρ_0 as a function of distance, from the transducer. Approximate transverse intensity distributions are plotted below this. (From Lemons, R. A., "Acoustic Microscopy by Mechanical Scanning," PhD thesis, E. L. Ginzton Laboratory, Stanford University, Stanford, CA,1975. With permission.)

The condition for a maximum or a minimum is given by the cosine term and is such that

$$z = \frac{4r_0^2 - n^2\lambda^2}{4n\lambda} \tag{6.38}$$

where $n = 1, 3, 5, 7, \ldots$ for a maximum and $n = 2, 4, 6, 8, \ldots$ for a minimum.

The final maximum occurs at $n = 1$ which corresponds to $z_F = a^2/\lambda$, which is called the Fresnel distance. Thus, the near field corresponds to $z < z_F$ and the far field to $z > z_F$. The near-field region is characterized by rapid interference maxima and minima, as shown in Figure 6.5. This makes sense physically as near the transducer a small shift along the axis leads to a relatively large shift in phase for the wavelet coming from a given surface element. This is not true in the far field where the phase shift is gradual and monotonic for all elements and the transducer acts more and more as a point-like source.

Finally, the variation of acoustic pressure in the transverse plane is also sketched in Figure 6.5. It is seen that the beam remains well collimated up to the Fresnel distance, although there are considerable intensity variations across the beam section. Beyond the Fresnel distance, in the far field, the beam widens, as expected, due to the increasing point source-like behavior.

6.2 SCATTERING

Scattering of acoustic waves by obstacles of various sorts is, as in most branches of physics, a highly developed and mathematically very sophisticated subject. As in other areas, the main results are relatively easy to present for the case where the wavelength is either much greater or much less than the characteristic dimension of the obstacle. The problem becomes much more difficult, often intractable, when the wavelength is of the order of this dimension. In this situation, we will content ourselves with an overview of scattering by a few simple objects.

In principle, as for the case of radiation, the scattered acoustic field can be determined from Huygens principle, adding the waves emitted from secondary sources over the surface of the

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scattering body, taking into account their relative amplitudes and phases. For a body of arbitrary size and shape, this problem is in general intractable. For scattering by simple objects, two approaches will be used to characterize the scattering: polar diagram and total scattered intensity as a function of frequency. The polar diagram is highly useful because it gives an immediate visual clue as to the intensity of sound scattered in a given direction. The total scattered intensity is displayed as a function of ka, where k is the wave number and a is a characteristic dimension of the scattering center. This graph is useful for identifying the various scattering regimes mentioned above. The two main examples to be discussed will be the cylinder and the sphere.

6.2.1 The Cylinder

We suppose a plane wave incident on a rigid cylinder of radius a in a direction perpendicular to the cylinder axis. In the geometrical optics limit $ka \gg 1$, the cylinder scatters as a geometrical obstacle in the back direction and scatters as interference between the incident wave and the forward scattered wave to produce a sharply defined geometrical shadow. This limit is more common in optics than acoustics due to the length scales involved, although it is easy attainable in an ultrasonic immersion tank.

The limit $ka \gtrsim 1$ requires detailed calculation, which has been carried out by Morse [1]. The main steps of the calculation are as follows:

1. Description of an incident plane pressure wave $(p, \dot{u}_{pr})$ in cylindrical coordinates.
2. Description of the outgoing wave $(p, \dot{u}_{sr})$ in terms of the same parameters (amplitude and phase) as in the first description above.
3. Calculate the amplitudes and phases of steps 1 and 2 to satisfy $\dot{u}_{pr} + \dot{u}_{sr} \equiv 0$ at $r = a$.
4. Calculate the scattered intensity as a function of angle, β a from the solutions of step 3.

For sufficiently short wavelengths, about one half of the intensity is scattered in the forward direction and the rest is scattered approximately uniformly over the remaining solid angle. This gives rise to a cardiod-type polar plot: It becomes more and more directive in the forward direction as the wavelength decreases, as shown in Figure 6.6. The total scattered intensity can also be calculated as a function of ka. For $ka \ll 1$, the scattered intensity rises rapidly to saturate for $ka \gg 1$.

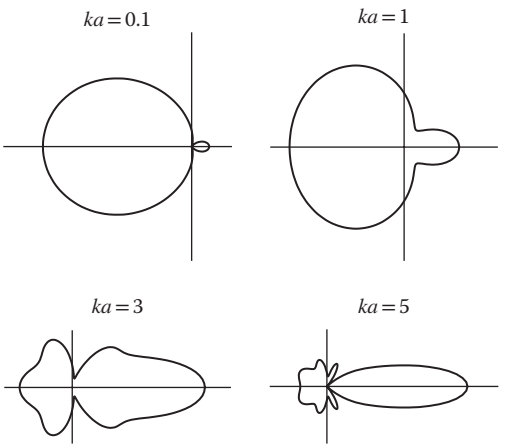

FIGURE 6.6 Polar diagrams (linear scale) for scattered radiation at wave number k from a rigid cylinder of radius a for $ka = 0.1$, 1, 3, and 5, respectively.

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6.2.2 The Sphere

The calculation follows the same lines as for the cylinder. The corresponding polar plots are shown in Figure 6.7, and the scattered intensity variation with ka is shown in Figure 6.8. The total scattered intensity as a function of ka will be discussed. The curve can be divided into three regions. For $ka \ll 1$, the curve approaches asymptotically to the classic Rayleigh formula $IS \sim 1/\lambda^4$ as it must be in the long wavelength limit. In the opposite, short wavelength limit, $ka \gg 1$, the reflection is mainly specular and the reflected intensity saturates. In the intermediate regime, the behavior is of a periodic nature due to the excitation of creeping or interface waves that travel around the curved surface of the obstacle at approximately the longitudinal sound velocity in the liquid.

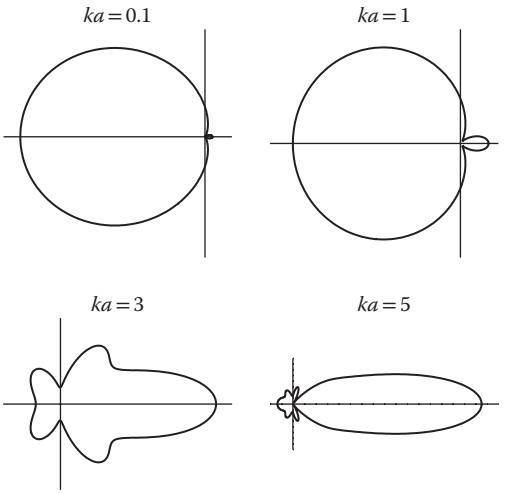

FIGURE 6.7 Polar diagrams (linear scale) for scattered radiation at wave number k from a rigid sphere of radius a for ka = 0.1, 1, 3, and 5, respectively.

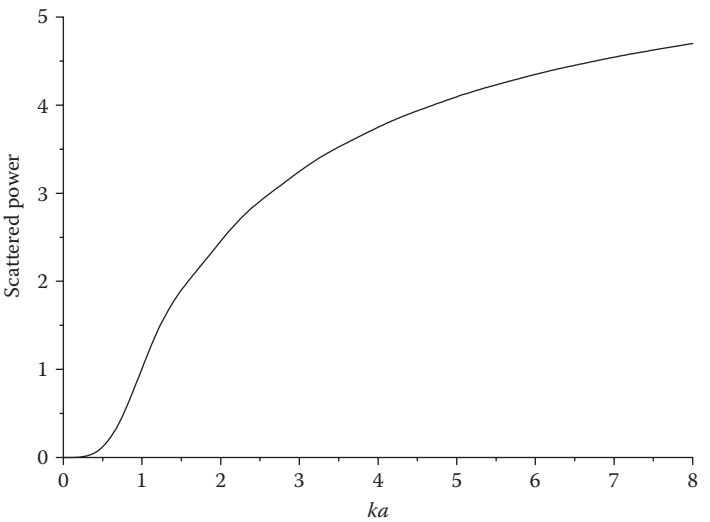

FIGURE 6.8 Scattering power of a sphere of radius a as a function of ka.

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The term scattering cross-section σ is commonly used to describe scattering problems; it is defined as the total scattered power divided by the incident intensity and represents the apparent area that blocks the wave. σ provides a convenient parameter to compare the scattering power of different forms of target. For example, for a sphere of radius a, $\sigma = 7/9\left(\pi a^2\right)\left(ka^4\right)$; it is seen that this form also incorporates the law for Rayleigh scattering.

6.3 FOCUSED ACOUSTIC WAVES

There are several levels of treatment for focused acoustic waves. The simplest, level 1, is to use geometrical optics or ray theory. For the spherically focused concave acoustic radiator to be considered in this section, level 1 immediately tells us that the acoustic energy is focused at the center of curvature. Level 2 takes into account diffraction, much in the same way that this has been handled for plane circular radiators in the previous sections of this chapter. This level demonstrates that the focal point is not an infinitesimal point but that it is spread out to the order of magnitude of the wavelength. This leads to the concept of point spread function and lateral resolution. The third level of sophistication recognizes that since the acoustic intensity is very high near the focus, nonlinear effects need to be taken into account. The main effect here is the generation of harmonics of the operating frequency in the focal region.

This book is limited to linear systems so level 3 will not be treated here although nonlinear effects in focusing will be discussed qualitatively in Chapter 17. Likewise, a full mathematical description of level 2 is beyond the scope of the book, and in any case has been provided in detail elsewhere by Kino [2], for example, whose general approach will be followed and summarized here. Given this we provide mainly a descriptive account of focused beams to a depth that will be sufficient to give an accurate description of acoustic lenses.

Rayleigh provided the first detailed treatment of the circular piston source described earlier, and these results will give a good first approximation for circular radiators, especially when the radius of curvature is much greater than the wavelength. Early treatments were provided by Williams [3] and O'Neil [4]. Lucas and Muir [5] reduced the surface integral over the radiator to a single integral and showed that within the Fresnel approximation the boundary conditions on the curved surface could be transformed to the plane of the baffle. Recently, a numerically convergent solution consistent with all limiting cases has been provided by Chen et al. [6].

Following Kino [2], we consider the focused spherical radiator shown in Figure 6.9. Using the results of Lucas and Muir, it is possible to consider the planar element AB as an effective source by taking into account the phase difference between a point on the surface of the spherical radiator and its corresponding point on the element AB using ray theory. Kino shows that this leads to the following expression for the displacement potential in the Fresnel approximation with $a^2 \ll z^2$:

$$\phi(r,z,\psi) = -\frac{u_0}{2\pi}\iint \frac{e^{-jk(R+R_1)}\cos^2\theta}{R\cos\theta_0}\, r\, dr\, d\psi \tag{6.39}$$

where u_0 is the amplitude of the radial displacement. Integrating over the azimuthal angle yields

$$\phi(r,z,0) = -e^{-jk(z+r^2/2z)}\frac{u_0}{z}\int_0^a J_0\left(\frac{krr}{z}\right) e^{\frac{-jkr^2}{2}\left(\frac{1}{z}-\frac{1}{z_0}\right)}\, r\, dr \tag{6.40}$$

where $J_0(x)$ is a Bessel function of the first kind of zero order. This result can be used to determine the displacement at the focus z_0

$$u_z(0,z_0) = \frac{j\pi a^2}{\lambda z_0} e^{-jkz_0} u_0 \tag{6.41}$$

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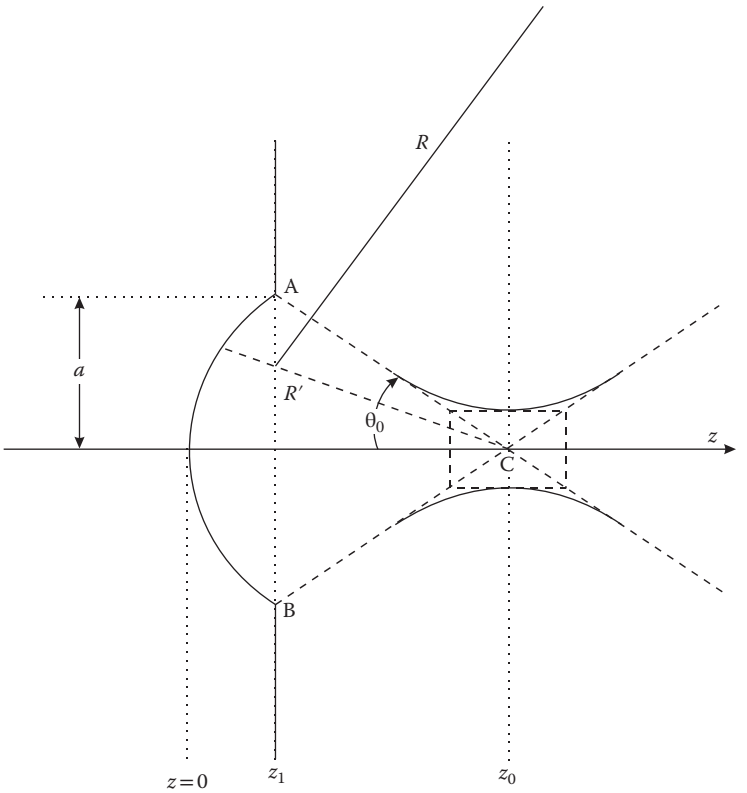

FIGURE 6.9 Focusing by a spherical radiator. The dotted cylindrical region around C gives the spatial resolution and depth of field.

for $kz_0 \gg 1$ and hence the beam intensity at the focus compared to that at the transducer.

$$\frac{I(0,z_0)}{I(0)} = \left(\frac{\pi a^2}{z_0\lambda}\right)^2 = \left(\frac{\pi}{S}\right)^2 \tag{6.42}$$

where $S = z_0\lambda/a^2$ is the Fresnel parameter. The lens will hence normally function in the regime $S < \pi$.

The lateral resolution can be determined by calculating the off-axis intensity at z_0. Equation 6.40 yields

$$\frac{I(r,z_0)}{I(0)} = \left(\frac{\pi a^2}{z_0\lambda}\right)^2 \text{jinc}^2\left(\frac{ra}{\lambda z_0}\right) \tag{6.43}$$

The main result here is that the lateral intensity varies as $\text{jinc}^2\,(ra/\lambda z_0)$, which is the same result as for a circular piston far from the source.

Equation 6.40 and its direct result, Equation 6.43, lead to quantitative criteria for the resolution.

1. Spatial resolution
   Using the Rayleigh criterion of resolution as in optics, the spatial resolution is given by the position of the first zero of the $\text{jinc}^2 x$ function

$$r_0(\text{zero}) = \frac{0.61\lambda}{\text{NA}} \tag{6.44}$$

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where NA = $\sin\theta_0$ is the numerical aperture.
The relative aperture or F number of the lens is given by

$$F = \frac{z_0}{2a} \tag{6.45}$$

2. Sidelobes
   The sidelobes are important in radiation patterns for plane transducers as has already been seen. Likewise for focused transducers they should be reduced as much as possible to improve signal discrimination. The first sidelobe for the spherical radiator occurs at the first secondary maximum of the $\text{jinc}^2 x$ function, at $kra/z_0 = 5.136$. It is 17.6 dB down in amplitude from the main lobe.
3. Depth of focus
   The axial variation of intensity can be determined from Equation 3.37, and with a suitable criterion, this can be used to determine the depth of focus. The simplest way to do this is to inscribe a cylinder in the focal region, as shown in Figure 6.8. From Equation 6.44, this gives a depth of focus along the z axis

$$d_z = 1.22\lambda \left(\frac{z_0}{a}\right)^2 \tag{6.46}$$

4. Phase change of π at the focus
   It has been shown in great detail by Born and Wolf [7] that there is a π phase change at the focus of three-dimensional focusing systems. This result also follows directly from Equation 6.40. An interesting discussion on this point is given in Royer and Dieulesaint [8]. The simple physical picture is as follows. A spherically converging wave front at the focus comes to a point and then exits the focus as a diverging spherical front. This corresponds to a reflection with respect to the origin (rotation by π), which corresponds to the π phase change.

6.4 RADIATION PRESSURE

Like all forms of radiation, a beam of acoustic energy will exert a force, or radiation pressure, on an object in its path. This phenomenon is important in the measurement of acoustic field and in the calibration of acoustic instruments such as hydrophones. The actual effect in laboratory or in field conditions can be quite complicated and depends on the specific configuration of the system under study. In what follows we give a simple treatment of an idealized case to bring out the basic principles involved. A good historical and tutorial account is given by Torr [9].

Consider the case of Figure 6.10 for a perfectly absorbing target. The standard construction for the energy flux is shown; during a time Δt, the energy contained within a cylinder of length $V_0\Delta t$ will attain the wall and be absorbed. For acoustic intensity I, the energy absorbed during time Δt is $IA\Delta t$. The wall will exert a force F against the wave and during time Δt will do work equal to $FV_0\Delta t$, which must be equal to the energy absorbed. Equating the two quantities and recognizing that by Newton's third law the wave will exert an equal and opposite force on the wall $F = p_r A$, we find for the radiation pressure

$$p_r = \frac{I}{V_0} \quad \text{(absorption)} \tag{6.47}$$

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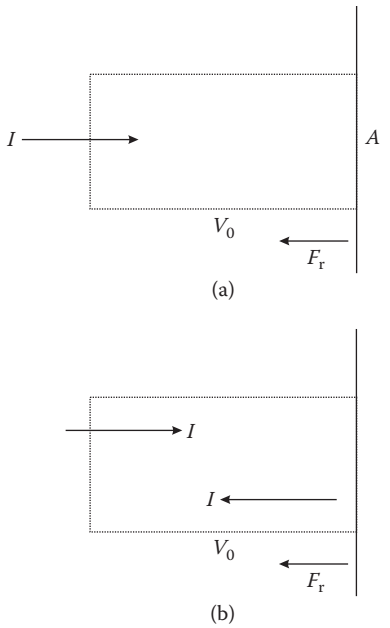

FIGURE 6.10 Geometry for acoustic radiation pressure. (a) Perfect absorber. (b) Perfect reflector.

For the case of a perfect reflector, the situation is similar to that for the pressure exerted by a perfect gas on the walls of the continuer. In that case, the calculation is usually made by putting the impulse, $F\Delta t$, equal to the change of momentum for particles inside the cylinder of Figure 6.10. For the case of absorption, the momentum to be absorbed is simply that of the incoming wave as calculated above. However, for the reflector, the direction of the momentum is reversed so that the impulse, or radiation pressure, is now determined by twice the modulus of the momentum of the incoming wave. Thus,

$$p_r = \frac{2I}{V_0} \quad \text{(reflection)} \tag{6.48}$$

In general, due to partial absorption, generation of different acoustic modes in the target, partial transmission in composite targets, and so on the actual radiation pressure will have a value somewhere between that of these two limiting cases.

6.5 DOPPLER EFFECT

A classic manifestation of the Doppler effect is that experienced unconsciously by every child watching a passing train. Here, the fixed observer (child) hears an apparent increase of frequency by the moving object (train) as it approaches, followed by a decrease as the train passes and then moves away. This Doppler frequency shift is important in ultrasonics, particularly for instrumentation for flow meters, medical applications, and oceanography. In these examples, any or all the source, medium, or receiver may be in movement.

The physical origin of the Doppler effect lies in the variation of the apparent wavelength. For the example of the moving source considered above, as the source emits spherical waves as it moves, the wave fronts in front of the source are scrunched together while those behind it become separated farther and farther apart, as shown in Figure 6.11. The corresponding effective changes in wavelength give rise to the observed frequency changes by the fixed observer.

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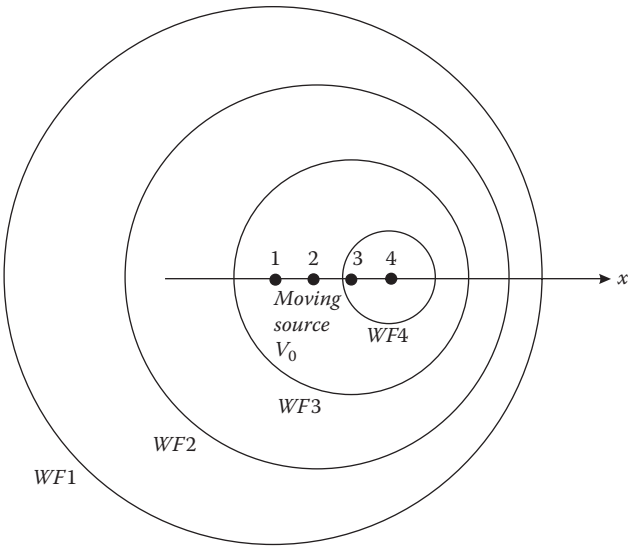

FIGURE 6.11 Crowding of wave fronts in front of a moving source leading to the Doppler shift. The situation is shown at four distinct source positions.

A quantitative estimate of the Doppler effect can be given as follows. We consider first motion along the axis for a moving source (V_S) and receiver (V_R) in a medium of sound velocity V_0, with emission by the source of a steady signal at frequency f_S. Due to the compressing of the wave fronts in front of the source, the wave front is shortened to

$$\lambda_a = \frac{(V_0 - V_S)}{f_s} \tag{6.49}$$

If the receiver is moving away from the source, it detects a frequency f_R

$$f_R = \frac{(V_0 - V_R)}{\lambda_a} \tag{6.50}$$

giving finally

$$f_R = f_S \frac{(V_0 - V_R)}{(V_0 - V_S)} \tag{6.51}$$

In a similar manner it can be shown that for a fixed source and receiver that radiates a frequency f_S toward a target with velocity V_T,

$$f_R = f_S \frac{(V_0 - V_T)}{(V_0 + V_T)} \tag{6.52}$$

More complete and rigorous demonstrations of the Doppler shift have been given in the literature, for example, Pierce [10].

Some of the applications of the Doppler shift will be mentioned here. Doppler methods for industrial flow meters for liquids and gases are numerous and are referred to in Chapter 16. Medical applications are numerous as this is the perfect technique for monitoring movement inside an opaque object. Blood flow, including fetal blood flow, is an obvious application. Others include movement of internal organic components (e.g., heart valves) and monitoring arteries for severity of athersclerosis.

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Oceanography instrumentation makes widespread use of Doppler, for example, for studying ocean layer dynamics using bubbles, plankton, and detritus as scattering centers for Doppler sonar. Monitoring movement of the sea surface and navigational aid are other applications. In the nonultrasonic world, Doppler radar is also an important application.

SUMMARY

Acoustic point source gives rise to spherical waves diverging from the point in question. The pressure amplitude of the acoustic wave varies as $1/r$.

Plane piston source is assumed to be uniformly excited across its face and to be enclosed in an infinite baffle such that acoustic energy is only radiated in the forward direction.

Fresnel distance from a circular plane piston source is given by $z_F = a^2/\lambda$. It is the position of the last intensity maximum along the axis going out from the transducer face.

Near field is that region between the Fresnel distance and the transducer face. It is characterized by strong variations in phase and amplitude of the acoustic wave.

Far field is that region far from the source and beyond the Fresnel distance. The amplitude varies as $1/r$ and the wave front approaches a plane wave the farther one goes from the source.

Scattering of acoustic waves can in principle be calculated from Huygens principle. The scattering amplitude is usually described by the scattering cross-section that represents the apparent area of the scattering object.

Focused acoustic radiator focuses the emitted acoustic waves at the center of curvature of the spherical radiator. The lateral intensity varies as $\text{jinc}^2\,(ra/\lambda z_0)$, which determines the spatial resolution at the focal point.

Radiation pressure of an acoustic wave is given by Equation 6.47 for an absorbing target and Equation 6.48 for a perfectly reflecting target.

Doppler effect is a change of observed frequency when source, target, or receiver are moving with respect to each other. The effect can be used to deduce the velocity of the target.

QUESTIONS

1. Draw the phasor diagram for the displacement, particle velocity, and pressure for the point source of Equation 6.13.
2. Draw the vector diagram for the specific acoustic resistance and reactance of the previous case.
3. Show that the specific acoustic reactance of a spherical wave is a maximum for $kr = 1$.
4. Calculate the average rate at which energy flows through a closed surface that surrounds a point source.
5. For a 5-mm-radius transducer, calculate the Fresnel distance in water as a function of frequency from 1 to 1000 MHz. Graph this result. Extend this result to a family of curves for liquids with sound velocities smaller and larger than that of water.
6. Sketch the radiation patterns for transducers of radius 1 and 10 mm into water at frequencies of 1 and 20 MHz. Explain the qualitative difference between the radiation patterns.
7. What are the implications for imaging if the side lobes of a focused beam compared to the main beam are 30 dB down and 3 dB down?
8. Reconcile Equations 5.33 and 6.24.
9. Calculate the formula for radiation pressure using the concept of momentum of a wave and Newton's second law.
10. For the case of a fixed receiver at angle θ to the motion of the source, show that Equation 6.51 becomes

$$f_R = f_S \frac{1}{1 - \frac{V_S \cos\theta}{V_0}}.$$

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